DEVISHI KESAR

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EDUCATION

German Cancer Research Center(DKFZ) ***Heidelberg, Germany***
Doctor of Natural Science *University of Heidelberg*
October 2026

Georgia Institute of Technology ***Atlanta, GA, USA***
Master in Bioinformatics
December 2020

Indraprastha Institute of Technology Delhi ***Delhi, India***
Bachelor of Technology (Honors) in Computer Science and Engineering
Minor in Computational Biology
August 2019

SKILLS

Programming and Computational Skills*:* Python, Bash, R, Java, Scala, Regex, SQL, MATLAB
Tools and Platforms: Git, Hive, Cytoscape, RStudio, PyMol, Databricks, AWS, Spark, Docker, Google Cloud Services

PUBLICATIONS

● Knoll, N., Masser, S., Bordas, B., Ebright, R. Y., Li, G., **Kesar, D.,** Destefanis, E., Kania, N., Rodriguez, D. J., Jen, J., Zagar, S. E., Mensah, C., Chen, Z., Moffitt, S. J., Enakireru, E. M., He, Y., Feng, B., Chokshi, M. K., Jin, C. Y., … Mulvaney, K. M. (2025). CRISPR-drug combinatorial screening identifies effective combination treatments for MTAP-deleted cancer. *Cancer Research*

● Shirole, N., **Kesar, D.,** Lee, Y., Goodale, A., Syamala, S., Kukreja, S., Li, R., Qiu, X., Yu, W., Goldman, S., Cejas, P., Long, H., Adelman, K., Doench, J., Sellers, W.R. and Kaelin, W., Sustaining Cyclin D1 Expression Renders Kidney Cancer Cells HIF2-Independent. (2025). *Cancer Discovery*

● Chaturantabut, S., Oliver, S., Frederick, D.T., Kim, J., Robinson, F.P., Sinopoli, A., Song, T.Y., He, Y., Chang, Y.C., Rodriguez, D.J., Chang, L., **Kesar, D.,** Ching, M., Dzvurumi, R., Atari, A., Tseng, Y.Y., Bardeesy, N., Sellers, W.R., Identification of potent biparatopic antibodies targeting FGFR2 fusion driven cholangiocarcinoma. (2025). *The Journal of Clinical Investigation*

● Chang, L., Jung, N.Y., Atari, A., Rodriguez, D.J., **Kesar, D.,** Song, T.Y., Rees, M.G., Ronan, M., Li, R., Ruiz, P., Chaturantabut, S., Ito, T., Tienen, L.M., Tseng, Y.Y., Roth, J., & Sellers, W.R. Systematic profiling of conditional pathway activation identifies context-dependent synthetic lethalities. (2023). *Nature Genetics*

●Li, R., Klingbeil, O., Monducci, D., Young, M. J., Rodriguez, D. J., Bayyat, Z., Dempster, J. M., **Kesar, D**., Yang, X.,Zamanighomi, M., Vakoc, C. R., Ito, T., & Sellers, W. R. Comparative optimization of combinatorial CRISPR screens. (2022). *Nature Communications*

● Mulvaney, K. M., Blomquist, C., Acharya, N., Li, R., Ranaghan, M. J., O'Keefe, M., Rodriguez, D. J., Young, M. J., **Kesar, D.,** Pal, D., Stokes, M., Nelson, A. J., Jain, S. S., Yang, A., Mullin-Bernstein, Z., Columbus, J., Bozal, F. K., Skepner, A., & Sellers, W. R. Molecular basis for substrate recruitment to the PRMT5 methylosome. (2021). *Molecular Cell.*

● Mason, M.J., Schinke, C., Eng, C.L.P. et al. Multiple Myeloma DREAM Challenge reveals epigenetic regulator PHF19 as marker of aggressive disease. *Leukemia (2020)* - **member of DREAM consortium**

● Yang, et al. Assessment of the Limits of Predictability of Protein and Phosphorylation Levels in Cancer. (2020), *Cell Systems* - **member of NCI-CPTAC-DREAM consortium**

RESEARCH EXPERIENCE

Pediatric glioma research lab, DKFZ
Heidelberg, Germany
Graduate student *October 2023 – Present*

- Processed ChIP-seq data to identify transcription factor binding patterns and histone modifications in pediatric gliomas
- Conducted transcription factor motif analysis to explore regulatory elements driving tumor-specific pathways
- Constructed a gene regulatory network, uncovering novel biomarkers associated with different glioma subtypes
- Leveraged machine learning and MOFA analysis to integrate multi-modal data, uncovering new relationships
- Performed single-cell analysis to trace biomarker expression and identify developmental blocks tied to tumor origin

Sellers Lab, Broad Institute of MIT and Harvard
Cambridge, MA, USA
Associate Computational Biologist *April 2021 – September 2023*

- Processed and analyzed data for CRISPR paralog redundancy screens to find actionable biomarkers
- Executed antibody docking and structure prediction for antibodies using modeling softwares
- Performed ChIPseq data analysis to look at TF motif enrichment and single nucleotide resolution cut profile
- Detected alternate splicing events helping suggest PBM interaction requirement for PRMT5-dependent splicing events
- Analyzed data from mass spectrometry experiments conducted to identify novel E3 substrates

Georgia Institute of Technology
Atlanta, GA, USA
Graduate Research Assistant *August 2019- December 2020*

- Benchmarked variant calling pipelines by comparison of performance of mappers and variant callers
- Cataloged genomic variants from Colombian populations and characterized their predicted functional impact
- Identified single variant disease associations using best practices from the American College of Medical Genetics
- Evaluated distribution of health-related genetic variants among populations from Colombia in light of their ancestry

EMD Serono
Billerica, MA, USA
Bioinformatics Summer Intern *May 2020 – July 2020*

- Developed Rshiny app to share isolated immune cell RNAseq from bulk breast cancer tumors with biological experts
- Discovered novel myeloid drug targets by leveraging differential expression and gene signature analysis
- Drove bioinformatics discovery by frequently presenting results to key stakeholders across multiple departments
- Identified actionable biomarkers supporting therapeutic remediation of immuno-oncology and autoimmune disease

POSTER PRESENTATIONS

- **Kesar, D.,** Keck, M.K., Joshi, P., Okonechnikov, K., Autry, R.J., T.W. Jones, D.T.W., Deciphering pediatric glioma subtypes—Super-enhancer dynamics and (epi)genomic insights into cell of origin, Biology of Genomes, CSHL 2025
- **Kesar, D.,** Keck, M.K., Joshi, P., Okonechnikov, K., Autry, R.J., T.W. Jones, D.T.W., Integrated (epi)genomic atlas for identification of biomarkers in pediatric gliomas, AACR 2025
- **Kesar, D.** Colombian Genome Diversity Project, GT@SC20 HPC Poster Show Proceedings
- **Kesar, D.** The discovery of context-specific paralog dependencies in cancer, Broad Retreat 2022

AWARDS

- Computational Biology Research Award – Spring 2020 and Fall 2020 – Georgia Tech
- Dean Academic Excellence Award – IIIT Delhi 2018
- RECOMB/ISCB Conference Travel Grant 2017

TEACHING EXPERIENCE

- Bulk RNA-seq computational analysis *Nov 2024-Present*
- Introduction to quantitative biology *Jan 2018-Apr 2018*

Instructor: Dr. Subhadip Raychaudhuri

LEADERSHIP AND VOLUNTEER EXPERIENCE

- Member, Social events team, DKFZ, Germany 2023-25
- Organizing team, Broad Retreat 2022, Broad Institute, USA 2021-22
- Volunteer, Charles river cleanup, The Charles River Conservancy, MA, USA, 2021-22
- Volunteer, GP Covid testing lab, Broad Institute, MA, USA, 2022

COURSES AND CERTIFICATIONS

● Research Management Training for Early-Stage-Researchers	Apr'25
● CRUK Children's Brain Tumor Centre of Excellence Summer School	Jul'24
● Cancer Genomics and Precision Oncology, online course by Harvard Medical School	Mar'22
● Immuno-oncology, online course by Harvard Medical School	Mar'22
● Precision medicine, online course by University of Geneva	Feb'21
● Starting single cell analysis, online course by EMBL-EBI	Jun'20
● Introduction to breast cancer, online course by Yale University	Jun'19
● Computer science and automation, Summer School | IISc Bangalore	Jul'18
● Summer school in Bioinformatics | NCBS	Jun'17